Soccer Matches

Updated business rules:

There can be multiple refs for a single match

A Player can play on many teams

- A match is between two Teams
- A Team can be matched against many other Teams
- A Ref can officiate many matches
- A Team has many Players
- *A match can be officiated by many Refs*
- *A Player can play on many teams*

F = {

MatchID -> DatePlayed, Location

MatchID + TeamID -> Score

RefID -> Phone, Address, City, State, Zip

MatchID + RefID ->

TeamID -> Name, Sponsor

PlayerID -> Name, Phone, DoB

TeamID + PlayerID ->

MatchID + PlayerID -> Goals, Penalties

}

Updated tables

Match(**MatchID (PK)**, DatePlayed, Location)

* Match_Ref(***MatchID (PK, FK)***, ***RefID (PK, FK)***)

Referee(**RefID (PK)**, Phone, Address, City, State, Zip)

Score(***MatchID (PK, FK)***, ***TeamID(PK,FK)***, Score)

Team(**TeamID (PK)**, Name, Sponsor)

* Team_Player(***TeamID (PK, FK)***, ***PlayerID (PK, FK)***)

Player(**PlayerID (PK)**, Name, Phone, DoB)

PlayerStats(***MatchID(PK,FK)***, ***PlayerID(PK,FK)***, Goals, Penalties)

Veterinary Visits

Updated business rules:

> They want to be able to record all the breeds for mixed-breed animals
>
> A visit can have multiple procedures performed
>
> They also want to begin tracking genealogical records of their animals by noting the mother and father.

- An Animal is assigned a single Category
- A Category can have many Animals
- An Animal may schedule many Visits
- A Visit is for a single Animal
- A Visit may have one Procedure
- A Procedure can be performed on many Visits
- A Vet can attend many Visits
- Each Visit is attended by a single Vet
- *An Animal can have one mother*
- *An Animal can have one father*
- *An Animal can have many Breeds*
- *A visit can have multiple procedures performed*

F = {
CategoryID -> Name, Description
AnimalID -> Name, DoB, Gender, Registered, Color, Photo, PetStore, *MotherID, FatherID BreedID ->*
VeterinaryID -> First_Name, Last_Name, Address, ZipCode, CityID, Phone
VisitID -> AnimalID, VetID, VisitDate
VisitID + *ProcedureID* -> Cost
ProcedureID -> Name, Description
}

Updated tables:
Category(**CategoryID (PK)**, Name, Descrption)
Animal(**AnimalID (PK)**, Name, DateBorn, Gender, Registered, Color, Photo, PetStore, MotherID, FatherID, *CategoryID (FK)*)
Breed(**Breed**, *AnimalID(FK)*)
Veterinarian(**VeterinaryID (PK)**, First_name, Last_name, Address, ZipCode, CityID, Phone)
AnimalVisit(**VisitID (PK)**, VisitDate, *AnimalID (FK)*, *VeterinaryID (FK)*)
Procedure(**ProcedureID(PK)**, Name, Description)
Visit(***VisitID(PK, FK)***, ***ProcedureID(PK, FK)***, Cost)

Dental Appointments

Update anomaly: If we update the name of the patiend in one record, we could forget to update it in another, creating appointments for two different names but the same patient.

Insert anomaly: If a new dentist joined the practice, there would be no place to put there information until they have appointments.

Delete anomaly: If an appointment is removed, the patient data could be lost as well if it's not stored in other appointments.

1NF (atomic values only): This table is already in 1NF as it does not have any compound values.

2NF
F = {
apptID -> staffNo, patientNo, date, time, surgeryNo
staffNo -> dentistName
patientNo -> patientName
}

3NF

Appointment(**apptID (PK)**, *staffNo (FK)*, *patientNo (FK*), date, time, surgeryNo)

Staff(**staffNo (PK)**, dentistName)

Patient(**patientNo (PK)**, patientName)

Library

State the normalization rule violations if any.

- Is the table in 1NF, why or why not?
    - No, there are multiple values for the Author field
- Is the table in 2NF, why or why not?
    - No, the book information is dependent on title, and not on any other parts of the key, and same for the customer information
- Is the table in 3NF, why or why not?
    - No, it is not in 1NF or 2NF
    - Also, the fine information is dependent on non-key attribute ReturnedDate

F = {
MID -> Name, Address, PhoneNum, Parent
ISBN -> Title, Authors
CheckoutID -> ISBN, BorrowDate, ReturnedDate, FineDue, FinePaid
}

Tables:
Member(**MID (PK)**, Name, Address, PhoneNum, Parent)
CheckOut(**CheckoutID (PK)**, *ISBN (FK)*, *MID (FK)*, BorrowDate, ReturnedDate, FineDue, FinePaid)
ISBN(**ISBN (PK)**, Title, Authors)